

JEE MAIN REVISION MASTER: ENVIRONMENTAL CHEMISTRY OUT OF SYLLABUS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 96 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Environmental Chemistry — Standard Calculations & Problem Archetypes	51 PYQs	Quantitative Analysis
Very High	Environmental Chemistry — Applied Cases, Reactions & Traps	41 PYQs	Applied Topics
High	Environmental Chemistry — Core Principles & Governing Laws	13 PYQs	Core Principle

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Environmental Chemistry — Standard Calculations & Problem Archetypes	$O(g) + O_2(g) \rightleftharpoons O_2(g) \xrightarrow{UV} O(g) + O(g), \quad ClONO_2$
Environmental Chemistry — Applied Cases, Reactions & Traps	$C_2Cl_2F_2 \rightleftharpoons C_2Cl_2 + 2F_2 \xrightarrow{h\nu} 2Cl + O_3 \rightarrow ClO + O_2$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Environmental Chemistry — Standard Calculations & Problem Archetypes (Very High Yield — 51 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Environmental Chemistry.

■ ■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2023 Evening Shift] Q#1 (MCQ): "The concentration of dissolved Oxygen in water for growth of fish should be more than $\times$ ppm and Biochemical Oxygen Demand in..." [Ans: N/A]
- [2023 Morning Shift] Q#1 (MCQ): "Which of the following is true about freons?..." [Ans: N/A]
- [2023 Morning Shift] Q#2 (MCQ): "Given below are two statements: Statement I : If BOD is 4 ppm and dissolved oxygen is 8 ppm, then it is a good quality water. Statement II : ..." [Ans: N/A]

3.2 Environmental Chemistry — Applied Cases, Reactions & Traps (Very High Yield — 41 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Environmental Chemistry.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2023 Morning Shift] Q#1 (MCQ): "The possibility of photochemical smog formation will be minimum at :..." [Ans: N/A]
- [2023 Evening Shift] Q#1 (MCQ): "Which of the following compounds is an example of Freon?..." [Ans: N/A]
- [2023 Evening Shift] Q#1 (MCQ): "Which of these reactions is not a part of breakdown of ozone in stratosphere?..." [Ans: N/A]

3.3 Environmental Chemistry — Core Principles & Governing Laws (High Yield — 13 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Environmental Chemistry.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Environmental Chemistry.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Environmental Chemistry PYQs.

Representative Source PYQs:

- [2023 Evening Shift] Q#3 (MCQ): "The water quality of a pond was analyzed and its BOD was found to be 4. The pond has :..." [Ans: N/A]
- [2022 Morning Shift] Q#3 (MCQ): "The eutrophication of water body results in :..." [Ans: N/A]
- [2021 Morning Shift] Q#3 (MCQ): "BOD values (in ppm) for clean water (A) and polluted water (B) are expected respectively :..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2023 (Online) 15th April Morning Shift (Q#1)

The possibility of photochemical smog formation will be minimum at :

- (A) Mumbai in May
- (B) Kolkata in October
- (C) New-Delhi in August (Summer)
- (D) Srinagar, Jammu and Kashmir in January

Correct Answer: None

Solution Analysis: The formation of photochemical smog depends on several factors, including sunlight, temperature, and the presence of pollutants such as nitrogen oxides and volatile organic compounds. Option (A) Mumbai in May may also have some possibility of photochemical smog formation because the temperature and sunlight are relatively high in May. Option (B) Kolkata in October may have some possibility of photochemical smog formation because the temperature and sunlight are still relatively high in October. Option (C) New-Delhi in August (Summer) is more likely to have photochemical smog formation because

Benchmark Problem 2: JEE Main 2023 (Online) 24th January Morning Shift (Q#1)

Which of the following is true about freons?

- (A) These are radicals of chlorine and chlorine monoxide
- (B) These are chemicals causing skin cancer
- (C) All radicals are called freons
- (D) These are chlorofluorocarbon compounds

Correct Answer: None

Solution Analysis: Freons are a family of chlorofluorocarbon (CFC) compounds, which are chemicals that contain chlorine, fluorine, and carbon atoms. They are used in a variety of applications, including refrigeration and air conditioning, as well as in aerosol spray cans.

Benchmark Problem 3: JEE Main 2022 (Online) 25th June Morning Shift (Q#3)

The eutrophication of water body results in :

- (A) loss of Biodiversity.
- (B) breakdown of organic matter.
- (C) increase in biodiversity.
- (D) decrease in BOD.

Correct Answer: None

Solution Analysis: Eutrophication is the process in which nutrient enriched water bodies support a dense plant population, which kills animal life by depriving it of oxygen and results in subsequent loss of biodiversity.